

Evaluating Unsupervised Clustering Strategies for Identifying Degradation-Related Chemical Domains in SEM-EDS Spectrum Imaging of Zinc-Soap Affected Paint Layers

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1. Introduction

The formation of metal soaps, or metal carboxylates, has emerged as a major topic of concern within the field of paintings conservation. In oil paintings, metal ions originating from pigment particles can react with free fatty acids in the drying oil binder to form metal carboxylate complexes.¹ These compounds may then aggregate and migrate within the paint layer, leading to the development of protrusions and other degradation features (**Figure 1**).^{1,2} Over time, such processes can result in cracking, delamination, and flaking of the painted surface, ultimately contributing to the loss or distortion of cultural heritage objects.^{3,4,5} The problem is particularly pronounced in modern oil paintings, where certain pigment formulations appear to promote metal soap formation more readily.⁶

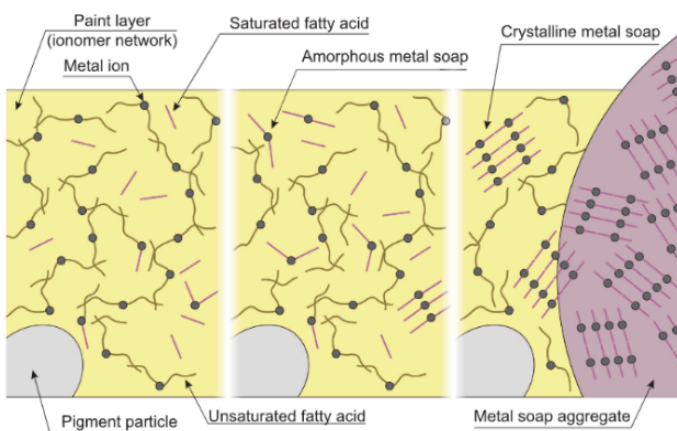


Figure 1. Schematic representation of the processes governing metal soap formation. From ¹.

Understanding the spatial distribution and chemical environment of metal soap formation is therefore an important objective for heritage science. Analytical techniques such as scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM-EDS) enable spatially resolved chemical characterization of paint layers and degradation products. SEM-EDS spectrum imaging produces hyper-spectral datasets in which each pixel contains a full X-ray spectrum, allowing elemental composition to be evaluated across the sample with high spatial

resolution. However, interpretation of these datasets often relies on qualitative visual inspection of elemental intensity maps, which can make it difficult to systematically identify subtle compositional relationships within chemically heterogeneous materials.

Alongside the increasing availability of complex analytical datasets, there has simultaneously been growing interest in the application of machine learning approaches within heritage science. Machine learning methods offer a framework for identifying patterns within high-dimensional data that may not be readily observable through traditional analysis.⁷ While supervised learning approaches have been widely explored, unsupervised methods are particularly well suited for exploratory analysis because they do not require labeled training data. Clustering algorithms can group chemically similar observations within a dataset, potentially revealing spatial domains associated with specific pigments, degradation products, or binder phases.

Despite this potential, the application of unsupervised machine learning methods to electron microscopy-derived spectroscopy datasets remains relatively limited.^{7,8} Applying clustering algorithms to SEM-EDS spectrum imaging data may provide a fruitful systematic approach for identifying chemically distinct regions within degraded paint layers and for exploring relationships between pigment composition, binder chemistry, and degradation products in the formation of metal soaps. Automated analysis of spectral imaging datasets therefore has the potential to complement traditional qualitative interpretation and provide new insights into the spatial organization of complex paint systems.

The objective of this study is to evaluate whether unsupervised clustering methods can identify chemically meaningful domains within SEM-EDS spectrum imaging data collected from a zinc-soap-affected oil paint test panel.⁹ Elemental intensity features extracted from hyperspectral EDS maps are used to construct multivariate feature vectors representing the composition of each pixel. After preprocessing and normalization, several clustering algorithms, including k-means clustering, Gaussian mixture models, and hierarchical clustering, are applied to these data. Cluster validation metrics are used to assess the stability and quality of the resulting segmentations, and spatial cluster maps are examined to evaluate whether the identified clusters correspond to chemically interpretable regions within the sample.

Through this analysis, the study aims to assess whether unsupervised learning can provide a reproducible framework for interpreting SEM-EDS spectrum imaging data and for identifying degradation-related chemical domains associated with metal soap formation in oil paint systems.

2. Methodology

2.1 Dataset and Data Acquisition

The datasets analyzed in this study are derived from SEM-EDS spectrum imaging measurements of an oil paint test panel designed to model zinc soap formation and degradation in painted systems.⁹ The test panel consists of a layered structure including a zinc oxide (ZnO)-based paint and cadmium red over a prepared ground. Following both accelerated and natural aging, protrusions associated with zinc carboxylate formation developed and were extracted for analysis in both aerial (surface) and cross-sectional configurations.

SEM-EDS spectrum imaging produced hyperspectral datasets in which each pixel contains a full X-ray spectrum. Two datasets are considered: an aerial map and a cross-sectional map of the protrusion region, providing spatially resolved elemental information.

Complementary point-based EDS spectra were also available and used as supplementary data for qualitative validation of clustering results, rather than as inputs to the clustering workflow.

2.2 Feature Extraction from Spectral Data

Element-specific feature maps were generated by integrating characteristic X-ray line intensities over predefined energy windows using tabulated values from the National Institute of Standards and Technology database.¹⁰ Selected features include carbon (C K α), oxygen (O K α), sulfur (S K α), chlorine (Cl K α), zinc (Zn K α).

Background subtraction was performed for chlorine using adjacent energy windows to remove continuum contributions, and negative values were clipped to zero. The resulting feature maps were stacked to form multi-channel feature cubes representing the spatial distribution of each element.

2.3 Masking and Data Transformation

Low-signal pixels were excluded using a conservative threshold based on the total X-ray counts per pixel, removing values below the 1st percentile to minimize noise contributions. Remaining pixel-wise feature vectors were flattened into a two-dimensional feature matrix.

A log transformation ($\log_{10}$) was applied to reduce intensity skew and compress dynamic range, followed by z-score normalization (zero mean and unit variance) to ensure comparable scaling across elemental channels prior to clustering.

2.4 Cluster Number Selection

The number of clusters was evaluated using a subset of the data and guided by the silhouette score, the Calinski-Harabasz index, and the Bayesian Information Criterion (BIC).^{11,12,13} These metrics were used heuristically, and a value of $k = 4$ was selected to balance interpretability and segmentation consistency across datasets and methods.

2.5 Unsupervised Clustering

Three clustering methods were applied to the standardized feature matrices: k-means, Gaussian mixture modeling (GMM), and hierarchical clustering (Ward linkage).^{14,15,16} Hierarchical clustering was performed on a subset of pixels due to computational constraints, with labels extended to the full dataset via nearest-centroid classification.

Cluster labels from GMM and hierarchical clustering were aligned to the k-means solution using centroid-based matching to enable direct comparison.

2.6 Spatial Reconstruction and Cluster Evaluation

Cluster assignments were mapped back to spatial coordinates to generate cluster maps for each dataset. Discrete colormaps were used to visualize cluster identity, and consistent label alignment ensured that corresponding clusters across methods could be compared directly.

Agreement between clustering methods was quantified using the adjusted Rand index (ARI).¹⁷ Pairwise comparisons were used to assess consistency between clustering approaches for each dataset. To interpret cluster chemistry, mean feature values were calculated for each cluster using log-transformed (but not re-standardized) feature intensities. These values were visualized as line plots to compare relative elemental contributions across clusters and between datasets.

2.8 Principal Component Analysis

Principal component analysis (PCA) was explored as a secondary analysis to assess whether the dominant structure of the data could be represented in a reduced-dimensionality space.¹⁸ PCA was applied to standardized feature matrices, and clustering results in PCA space were compared qualitatively to those obtained in the original feature space. This analysis was used for exploratory visualization rather than as a primary analytical method.

3. Results & Discussion

3.1 Spatial Clustering of Aerial and Cross-Section Datasets

Spatially coherent and consistent clustering was observed across all three methods (k-means, GMM, and hierarchical clustering) and for both the aerial and cross-section datasets (**Figure 2**). In each case, clustering forms contiguous regions that correspond to distinct compositional domains, indicating that the underlying chemical structure of the data is well captured by unsupervised approaches.

While the overall segmentation is consistent between methods, subtle differences are apparent. In particular, GMM produces comparatively smoother transitions between clusters, with less sharply defined boundaries between large regions. This is most evident in the broader spatial distribution of cluster 3, which appears as a diffuse, low-intensity presence throughout the sample rather than being confined exclusively to discrete regions. In contrast, k-means and hierarchical clustering yield more sharply delineated cluster boundaries, resulting in more clearly segmented domains.

The strong agreement in spatial structure across clustering algorithms suggests that the identified clusters correspond to chemically distinct regions that are robust to the choice of method. This consistency supports the interpretation that the segmentation reflects intrinsic compositional differences within the sample, rather than artifacts of a particular clustering approach.

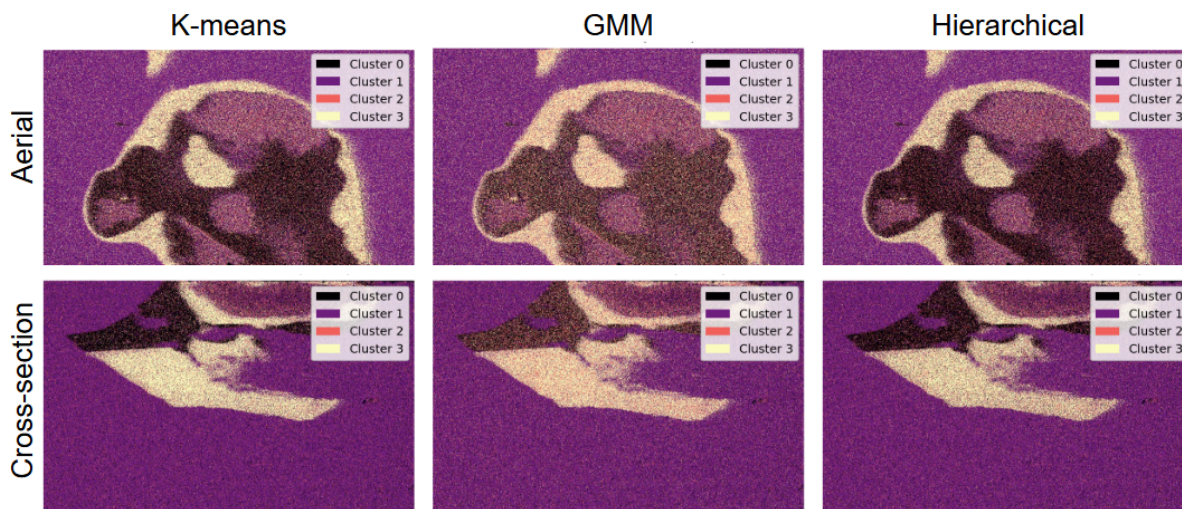


Figure 2. Spatial clustering results for aerial (top row) and cross-section (bottom row) SEM-EDS datasets using k-means, Gaussian mixture modeling (GMM), and hierarchical clustering. All methods produce spatially coherent cluster maps with consistent segmentation of compositional domains. GMM exhibits smoother transitions between clusters, while k-means and hierarchical clustering yield more sharply defined boundaries. Cluster labels are aligned across methods based on the k-means solution for direct comparison.

3.2 Agreement Between Clustering Methods and Datasets

High agreement was observed between clustering methods for both the aerial and cross-section datasets, with adjusted Rand index (ARI) values ranging from 0.85 to 0.96 (**Figure 3**). Across both datasets, k-means and hierarchical clustering exhibited the strongest agreement, with ARI values of 0.94 for the aerial dataset and 0.96 for the cross-section dataset.

Gaussian mixture modeling (GMM) showed slightly lower agreement with the other methods, with ARI values of 0.85 (k-means) and 0.86 (hierarchical) for the aerial dataset, and 0.90 (k-means) and 0.93 (hierarchical) for the cross-section dataset. Despite these differences, all pairwise comparisons indicate strong consistency in cluster assignments across methods, again supporting that the identified clusters correspond to chemically distinct regions that are robust to the choice of method.

Overall, the cross-section dataset demonstrates marginally higher agreement between clustering methods compared to the aerial dataset. This suggests that the cross-sectional geometry may provide more clearly defined compositional domains, leading to more consistent segmentation across algorithms.

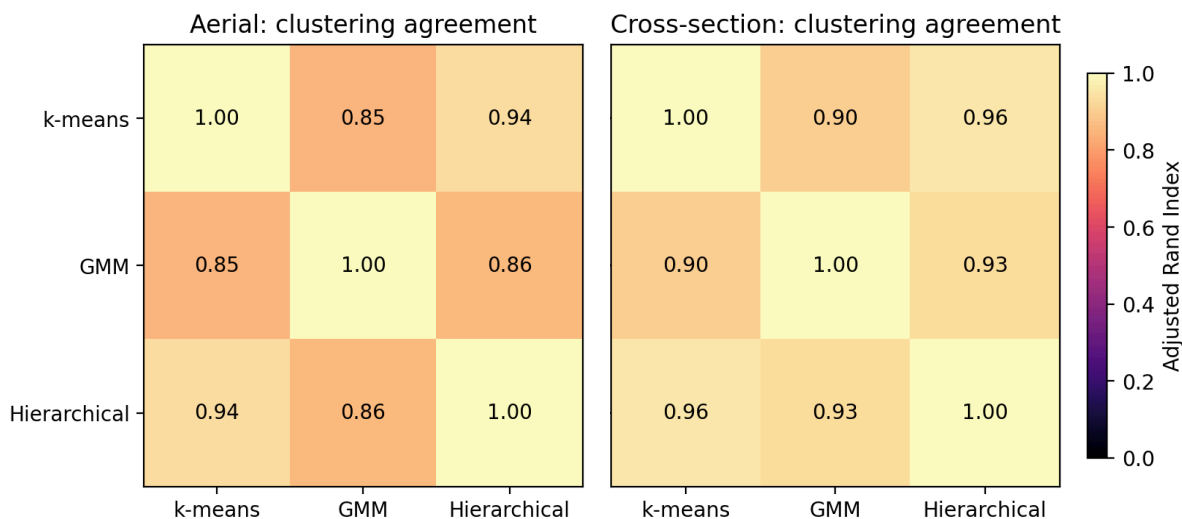


Figure 3. Adjusted Rand index (ARI) comparison between clustering methods for aerial (left) and cross-section (right) datasets. All methods show strong agreement (ARI = 0.85-0.96), with the highest consistency observed between k-means and hierarchical clustering. The cross-section dataset exhibits slightly higher agreement overall, indicating more well-defined compositional domains relative to the aerial dataset.

3.3 Cluster Chemistry

To further assess agreement between clustering methods and datasets, the chemical composition of each cluster was evaluated by comparing mean feature intensities, representing the relative contribution of each element to a given cluster.

Across all analyses, four chemically distinct regions are consistently identified: a zinc-rich region, a sulfur-rich region, a chlorine-rich region, and a matrix-dominated region. The zinc-rich cluster captures the majority of the Zn signal and is clearly separated from other regions. Similarly, the sulfur- and chlorine-rich clusters are defined by elevated S and Cl contributions, respectively.

The matrix-dominated cluster is characterized by comparatively higher carbon and oxygen signals, which are present in all clusters but do not dominate in the more chemically distinct regions. In contrast to the Zn-, S-, and Cl-rich clusters, this region lacks a single defining elemental peak and instead reflects areas where the underlying matrix composition is most prominent.

Consistent with previous results, strong agreement is observed between the aerial and cross-section datasets, with both geometries identifying the same set of chemically distinct regions (**Figure 4**). While Figure 4 presents this comparison for k-means clustering, these trends are representative of all clustering methods, as demonstrated by the high spatial consistency and agreement metrics discussed in Sections 3.1 and 3.2.

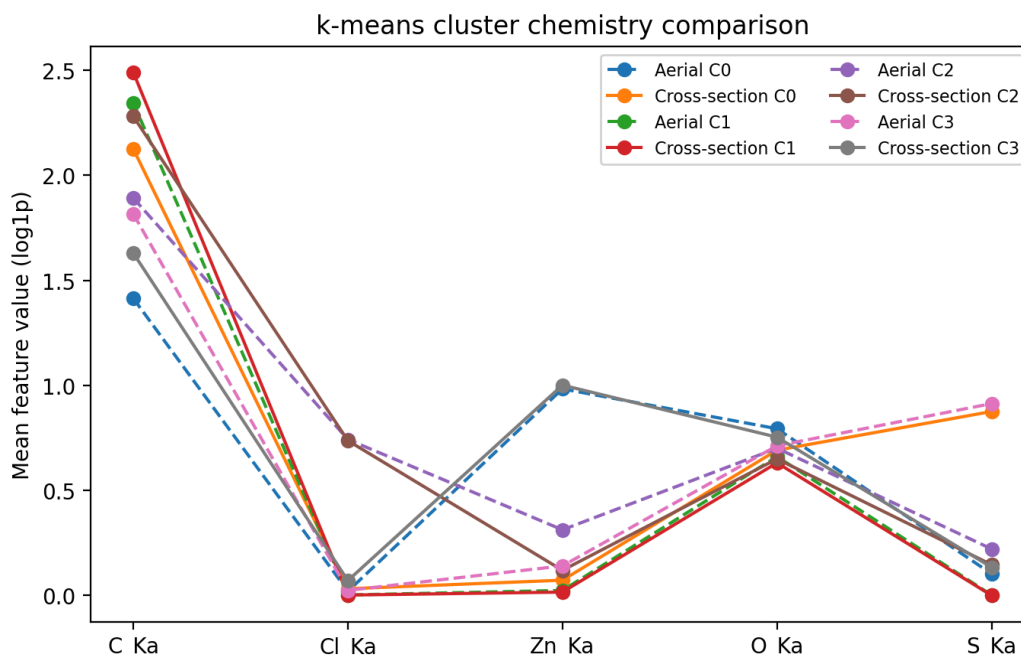


Figure 4. Comparison of mean feature intensities (log1p-transformed) for k-means clusters in aerial and cross-section datasets. Clusters exhibit consistent chemical signatures across both geometries, including zinc-rich, sulfur-rich, chlorine-rich, and matrix-dominated regions. Similar trends are observed across all clustering methods.

3.5 PCA-Based Clustering Comparison

As supplementary analysis, principal component analysis (PCA) was applied prior to k-means clustering to evaluate whether dimensionality reduction alters the resulting segmentation. Clustering performed in PCA-transformed space shows strong agreement with the original feature-based approach for both aerial and cross-section datasets, as observed in the generated spatial cluster maps (**Figure 5**). Spatial cluster maps from PCA-reduced data exhibit nearly identical morphology and boundary definition compared to those obtained using the original feature matrix. Similarly, cluster-wise chemical signatures remain consistent with the same zinc-rich, sulfur-rich, chlorine-rich and matrix-dominated regions across both approaches.

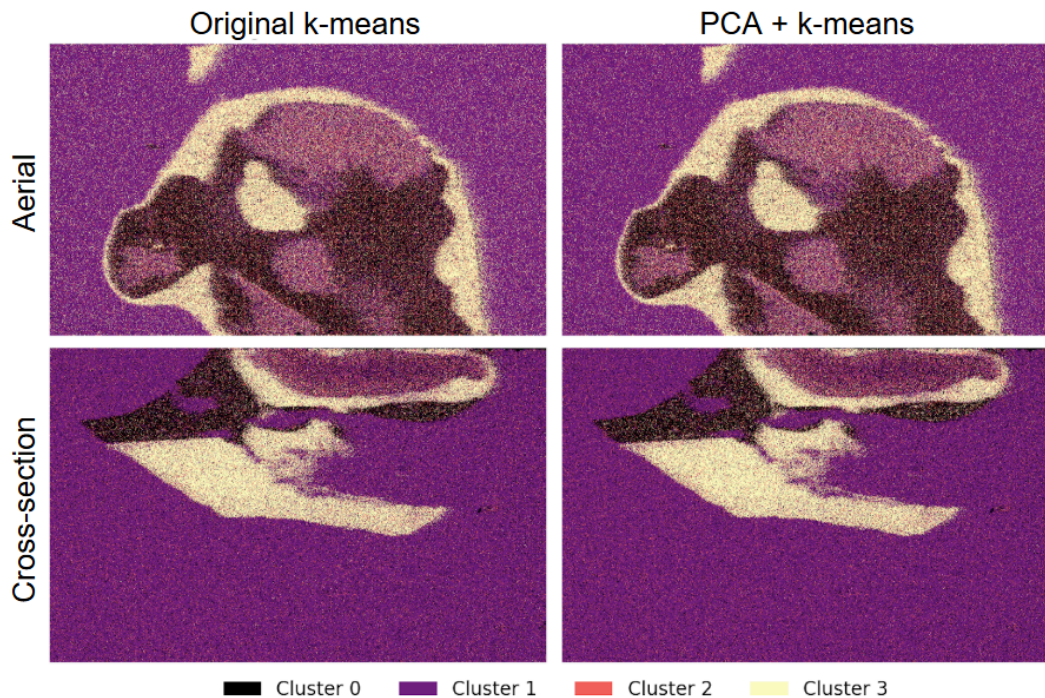


Figure 5. Comparison of k-means clustering performed on the original feature matrix (left) and PCA-transformed data (right) for aerial (top) and cross-section (bottom) datasets. Nearly identical spatial segmentation is observed in both cases, indicating that dimensionality reduction preserves the primary compositional structure of the data.

Visualization of the PCA-transformed feature space further supports this agreement, with data points forming structured distributions that correspond to the k-means cluster assignments (**Figure 6**). These results indicate that the dominant compositional structure of the dataset is preserved under dimensionality reduction.

Given the relatively low dimensionality of the present feature matrix (five elemental features), the minimal impact of PCA is expected. However, this result demonstrates that PCA can be incorporated without loss of interpretability, suggesting its utility for future applications involving higher-dimensional datasets or more complex compositional systems.

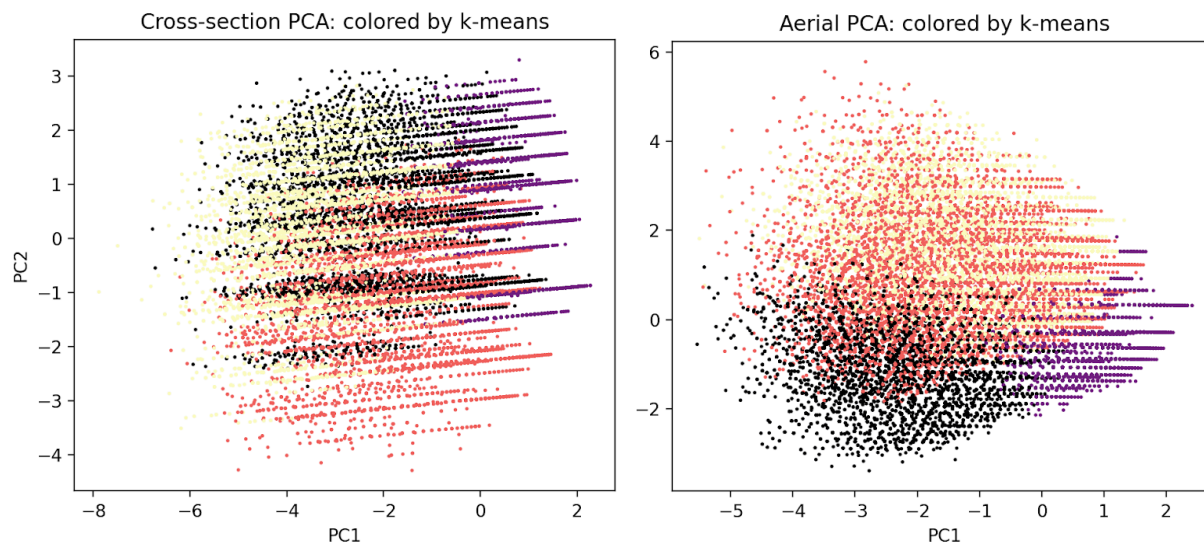


Figure 6. Principal component analysis (PCA projections of aerial (left) and cross-section (right) datasets, colored by k-means cluster assignments. Distinct clustering structure is preserved in reduced-dimensional space, supporting consistency between PCA-based and original clustering results.

3.6 Point-Scan Spectral Comparison

Point-scan SEM-EDS spectra provide independent validation of the compositional differences identified in the clustering analysis (**Figure 7**). Spectra acquired from regions of interest show clear distinctions between zinc-rich and cadmium-rich compositions. Zinc-rich spectra are characterized by strong Zn L and Zn K emission, while cadmium-rich spectra exhibit prominent Cd L peaks accompanied by sulfur signals.¹⁰

Within the mixed “White and Red” dataset, spectra can be reliably separated into zinc-white-like and cadmium-red-like groups based on these characteristic features. This separation is consistent with the chemically distinct regions identified through unsupervised clustering.

Additional minor peaks are observed in the cadmium-rich spectra and are attributed to silicon escape peaks associated with high-energy X-ray interactions in the detector. Such artifacts are well documented in energy-dispersive X-ray spectroscopy and arise from incomplete energy deposition within the detector system.¹⁹

Overall, the strong correspondence between point-scan spectral signatures and cluster-derived chemical trends supports the validity of the clustering approach and confirms that the identified clusters reflect meaningful compositional variation.

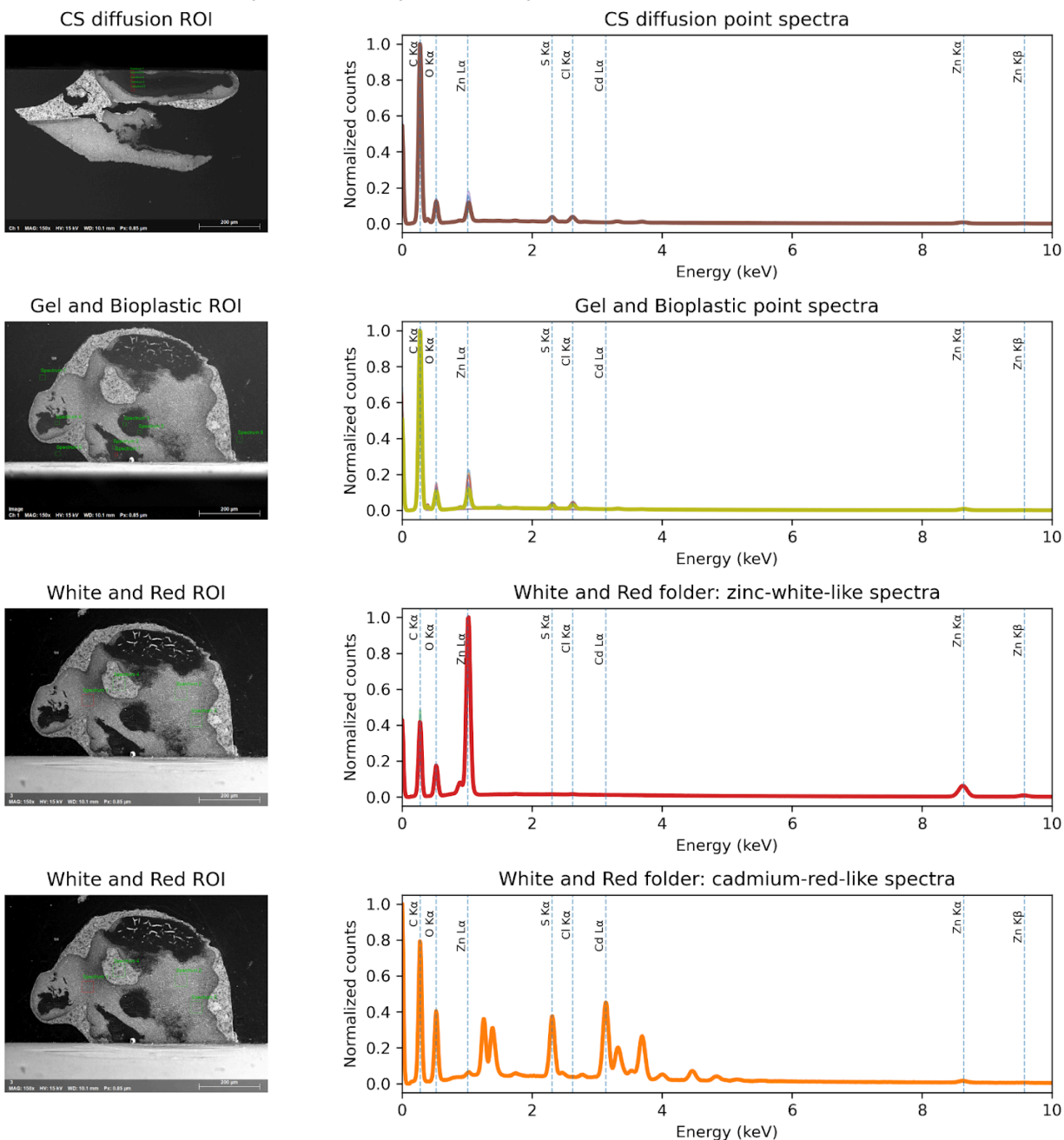


Figure 7. Representative SEM-EDS point spectra (right) with corresponding regions of interest (ROIs) (left). Spectra from the CS diffusion and gel/bioplastic samples show consistent elemental features, while the mixed “White and Red” dataset is separated into zinc-white-like and cadmium-red-like spectra based on characteristic peak intensities. Zinc-rich spectra exhibit strong Zn L and Zn K peaks, whereas cadmium-rich-spectra display prominent Cd L and associated sulfur signals. Vertical dashed lines indicate reference positions of characteristic X-ray emission lines.

4. Conclusion

Unsupervised clustering of SEM-EDS spectral imaging data was successfully implemented to identify chemically distinct regions within both aerial and cross-section datasets and across clustering algorithms. Across all three clustering methods, k-means, GMM, and hierarchical

clustering, spatially coherent and reproducible segmentation was observed, indicating that the dominant structure of the data is robust to algorithm choice.

Cluster specific chemical signatures, derived from elemental feature intensities, revealed clear distinctions between zinc-rich, sulfur- or chlorine-associated regions. These trends were consistent across datasets, while differences in cluster distribution highlighted variations between surface and subsurface composition.

Independent validation using point-scan EDS spectra further supported the clustering results. Spectra from regions of interest exhibited characteristic features consistent with the clusters identified during unsupervised clustering, including zinc-rich and sulfur-rich compositions, confirming that the unsupervised segmentation reflects meaningful chemical variation rather than purely statistical separation.

Together these results demonstrate that relatively simple unsupervised methods, when applied to deliberately preprocessed spectral imaging data, can provide a robust and scalable framework for resolving computational heterogeneity in complex cultural heritage materials and for enabling more systematic interpretation of spectral imaging datasets.

5. Future Directions

Several avenues exist to extend and refine this approach. Incorporating richer spectral representations, such as full-spectrum analysis or peak fitting methods, would improve sensitivity to subtle compositional differences that may be obscured by fixed energy window integration. Further improvements in feature extraction, including adaptive energy window selection and more rigorous background subtraction, could reduce overlap between elemental signals, particularly for low-energy peaks (e.g., C and O) and for elements with known spectral interference, such as Zn L and neighboring transitions.

In addition, incorporating spatial context into the clustering framework, either through spatially aware algorithms or neighborhood-based regularization, may improve segmentation of diffuse or transitional regions that are not well captured by purely feature-based approaches. Expanding the application of this methodology to a broader range of samples, including historically aged paintings with well-characterized conservation histories, would provide stronger validation and enable more direct connections between clustering results and degradation mechanisms such as metal soap formation.

Finally, extending this framework to higher-dimensional and multimodal datasets, such as those combining SEM-EDS with FTIR or Raman imaging, represents a promising direction for integrating complementary chemical information within a unified unsupervised learning approach.

6. References

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